THE COMPLEXITY OF CYCLE MIXING

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Abstract. This includes parts of the paper on complexity of mixing and reconfiguration.

1. Introduction

In this paper, all graphs are finite and undirected, but may have loops. A graph is called *reflexive* if all loops are reflexive, and *irreflexive* if none are. A *homomorphism* $\phi : G \to H$ from a graph G to a graph H is an edge preserving vertex map. Given graphs G and H, $\mathrm{Hom}(G, H)$ is the set of homomorphisms from G to H. The Hom-graph $\mathbf{Hom}(G, H)$ is the graph on the vertex set $\mathrm{Hom}(G, H)$ in which two vertices $\phi, \psi \in \mathrm{Hom}(G, H)$ are adjacent if for all edges uv of G, $\phi(u)\psi(v)$ is an edge of H.

This leads to the following problem of H-reconfiguration (or H-recolouring[1]) which has much interest lately [**?**].

Problem: $\mathrm{Recon}(H)$
Instance: A graph G and $f, g \in \mathrm{Hom}(G, H)$.
Question: Is there a path in $\mathbf{Hom}(G, H)$ between f and g?

This is clearly related to the problem H-mixing, which is the main problem of this paper, and which has also been addressed lately [**?**, **?**].

Problem: $\mathrm{Mix}(H)$
Instance: A graph G.
Question: Is $\mathbf{Hom}(G, H)$ connected?

As a given path between homomorphisms can be verified one edge at a time in polynomial space, both problems are in the complexity class PSPACE. In [**?**] there are many examples of graphs for which both problems are PSPACE-complete. In [**?**] the analogous problems were addressed for binary relational structures H. In this setting the H-reconfiguration problem was shown to exhibit dichotomy. Depending on H, $\mathrm{Recon}(H)$ is either polynomial time solvable, or PSPACE-complete. Once $\mathrm{Recon}(H)$ is known to by polynomial time solvable for a given structure H, the problem $\mathrm{Mix}(H)$ drops into the complexity class coNP, as we now have a polynomial certificate of the disconnectedness of $\mathbf{Hom}(G, H)$. In [**?**], they went on to conjecture a trichotomy for $\mathrm{Mix}(H)$ for binary relational structures H: depending on H, $\mathrm{Mix}(H)$ is polynomial time solvable, coNP-complete, or PSPACE-complete. They

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[1]The two problems, defined differently, coincide for graph, but not for digraphs. The distinction is addressed in [**?**].

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conjecture a classification for the trichotomy, and gave strong evidence supporting it.

The expectation is that with graphs H we should have a similar complexity breakdown– a dichotomy between P and PSPACE-complete for $\mathrm{Recon}(H)$, and a trichotomy between P, coNP-complete, and PSPACE-complete for $\mathrm{Mix}(H)$. The graphs for which the problems are PSPACE-complete should coincide. Though it seems premature to suggest a classification of the graphs H for which these problems fall into the respective complexity classes, some discussion of this is given in [**?**] for $\mathrm{Recon}(H)$ for both reflexive and irreflexive graphs H.

There is not much known about the complexity of mixing for graphs. For irreflexive graphs, it was shown in [**?**] that $\mathrm{Mix}(K_3)$ is coNP-complete, and recently in [**?**] this was extended to all odd cycles. Nothing yet has been proved for irreflexive graphs, but there is one basic result from the literature that immediately suggests a topological approach.

Indeed, from [**?**], we have that if H' dismantles to H, then $\mathbf{Hom}(G, H')$ has the same number of components as $\mathbf{Hom}(G, H)$ for all G. So the complexity of $\mathrm{Mix}(H')$ is the same as that of $\mathrm{Mix}(H)$. From this we conclude the following. This shows that $\mathrm{Mix}(H)$ is trivial for any (reflexive) dismantlable graph H.

In the case of the reconfiguration problem one finds a similarity between results for reflexive graphs and irreflexive graphs, and the topological techniques we use are 'cleaner' in the reflexive setting, so we start there. The reflexive analogue of the fact that $\mathrm{Mix}(C)$ is coNP-complete for irreflexive odd cycles of girth at least 5 is that $\mathrm{Mix}(C)$ is coNP-complete for reflexive cycles of girth at least 4. Our goal was to prove this, but in doing so we found the proof could be extended, with not too much work to all triangle free reflexive graphs.

In [**?**], we adapted to results of Wrochna [**?**] for irreflexive graphs to reflexive graphs, and showed that $\mathrm{Recon}(H)$ is polynomial time solvable (for reflexive instances) for any triangle-free reflexive graph H. As discussed above, this yields the following corollary.

?⟨fact:coNP⟩? **Fact 1.1.** *For any triangle-free reflexive graph H, the problem* $\mathrm{Mix}(H)$ *is in the complexity class* coNP*.*

In this paper, we prove completeness with the following theorem.

⟨thm:full⟩ **Theorem 1.2.** *For any triangle-free reflexive graph H, the problem* $\mathrm{Mix}(H)$ *is* coNP*-complete.*

The proof depends on results from [**?**] and exploits the same connection to topology that is central to the algorithms there and in [**?**]. This lends support to the notion that topology may factor into the trichotomy classification. We expect the proofs and results of this paper should yield somewhat messier irreflexive versions depending on [**?**] rather than on [**?**].

2. Outline of Proof and of Main Construction

⟨sect:outline⟩ One of the essential features in many papers on recolouring and homomorphism reconfiguration is the simple fact that when mapping a cycle B to another smaller cycle C of girth at least 4, (or 5 if B is not reflexive) the number of times that B winds around C is invariant under reconfiguration. In [**?**], Wrochna, to great effect, extended this idea and showed that for a cycle C mapped to graph H the

homotopy type of the image is invariant under reconfiguration. We did the same for reflexive graphs in [**?**]. In this paper we use the slighly coarser, but computationally simpler topological invariant of homology. We define homology properly in Section 3, but for the purposes of this outline, one unfamiliar with the concept benefits by thinking of H as a cycle, and the homology of a cycle mapping to H as counting the number of times it winds around H; having trivial homology means it does not wind.

As H is reflexive,there are always constant maps in $\mathrm{Hom}(G, H)$, and as H is connected, it is easy to see that these are always in the same component of $\mathrm{Hom}(G, H)$. So to decide if $\mathrm{Hom}(G, H)$ is connected, one must decide if every H-colouring of G reconfigures to a constant map. In [**?**] we showed that an H-colouring of G reconfigures to a constant map if and only if it has trivial homotopy (see Corollary 4.4 below), and trivial homotopy is implied by trivial homology (see Lemma 4.5). So deciding if all H-colourings of G have trivial homology is a subproblem of $\mathrm{Mix}(H)$. Calling an H-colouring of G *flat* if it has trivial homology, we reduce our task to showing NP-completemness of the problem $\mathrm{NonFlat}(H)$ of deciding if there is a non-flat H-colouring of an instance G (see Corollary **??**).

To prove this, we will reduce 3-colouring to $\mathrm{NonFlat}(H)$ via a gadget construction. Given an (irreflexive) instance G of 3-colouring, we construct an instance G^* of $\mathrm{NonFlat}(H)$ such that G has a 3-colouring if and only if G^* has a non-flat homomorphism to H. Now, we give a overview of the construction.

The (categorical) product $A \times B$ of two graphs A and B is the graph with vertex set $V(A) \times V(B)$, in which $(a, b) \sim (a', b')$ if $a \sim a'$ in A and $b \sim b'$ in B. For a vertex b of B, $A \times b$ is the subgraph induced on the vertices $\{(a, b) \mid a \in A\}$. If b has a loop, then this is an isomorphic copy of A. The subgraph $a \times B$ is defined similarly.

Where P_ℓ is the reflexive path on vertices $0, \ldots, \ell$, it is well known, and easy to show, that the existence of a path P_ℓ from ϕ to ψ in $\mathrm{Hom}(G, C)$ is equivalent to the existence of a homomorphism from $P_\ell \times G$ to C that restricts on $0 \times G$ to ϕ and on $\ell \times G$ to ψ. For a cycle C, we refer to $P_\ell \times C$ as a *path-of-cycles* and to each copy $i \times C$ of C a *slice* of the path.

In Fact 3.4, we observe that under a homomorphism to H, all slices have the same homology. This property of paths-of-cycles is the main building block in our construction of G^*. Let C be a smallest cycle in H. G^* will contain a special copy Z of C, and for every vertex v of G, three more copies A_v^1, A_v^2 and A_v^3 of C. Under any homomorphism to H, each of these either have trivial homology, or that of a copy of C in H.

Using a gadget S_3 made of paths-of-cycles we will ensure that under a homomorphism $\phi : G^* \to H$, if Z has non-trivial homology, then exactly one of the A_v^i for each v have the same homology as Z, and the other two have trivial homology. For every edge uv of G we use another path-of-cycles gadget S_2 to ensure that not both A_v^i and A_u^i get non-trivial homology. From this it will not be hard to show that if there is a non-flat homomorphism $\phi : G^* \to C$; one such that Z has non-trivial homology, then we get a 3-colouring of G by mapping v to the i such that A_v^i has non-trivial homology.

The construction as described above is not so difficult. The gadgets are constructed in Section 5 and the rest of the construction is done in Section 6. However, we also have to show that if there is no 3-colouring of G, then all homomorphism of

G^* to H are flat. It is not difficult to show that if there is no 3-colouring of G, then any homomorphism of G^* to H has trivial homology the slices of our path-of-cycles gadgets. This does not yet ensure that all H-colourings of G^* are flat. There may be a lot of other 'non-gadget' cycles in G^* that also must have trivial homology. This complicates the whole construction.

To deal with this we define a subgraph T of G^* that, in a way, contains all of these non-gadget cycles. We will 'plug' each cycle in T by triangulating it to a new central vertex. So that this does not break the necessary properties of G^*, we must build it, and the gadgets, so that we know the possible colourings on the vertices that will be in T. So our gadget construction mentions a subgraph T. The proof that that the cycles in T that we plug in this way are all the cycles that have to be plugged takes considerable work. This proof is done in Section 7 and the actual 'plugging' construction is then finished in Section 8.

3. The homology of a graph

⟨sect:homology⟩ We recall the basic definitions of simplicial homology with coefficients in $\mathbb{Z}$. This is basically Section 2.1 of [**?**] applied to the clique-complex of a graph G.

In the *clique-complex* $\mathbf{G}$ of a graph G, an *n-simplex* is an ordered $(n+1)$-clique of G, where as we are reflexive, we consider multisets $[v, v, v]$, or $[u, v, u]$ if $u \sim v$, as 3-cliques. The *chain group* $C_n(\mathbf{G})$ of G is the (additive) group generated as a $\mathbb{Z}$-module over the set of n-simplices of $\mathbf{G}$. Its elements are n-chains– sets of n-simplices. Between these groups we have the boundary maps $\delta_n : C_n(\mathbf{G}) \to C_{n-1}(\mathbf{G})$ defined on n-simplices by

$$\delta_n([v_0, v_1, \ldots, v_n]) \sum_{i=1}^{n} (-1)^i [v_0, v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n],$$

and extended to the whole group additively. (Note that $C_{-1}(\mathbf{G})$ is taken to be 0, and δ_0 the 0-map.) For example, assume that G has two 3-cliques, $\{a, b, c\}$ and $\{a, d, b\}$. The sum, $[a, b, c] + [a, d, b]$ say, is a 2-chain. Its boundary is

$$\begin{aligned} \delta([a, b, c] + [a, d, b]) &= [b, c] - [a, c] + [a, b] + [d, b] - [a, b] + [a, d] \\ &= [a, d] + [d, b] + [b, c] - [a, c]. \end{aligned}$$

The homology groups are defined using the boundary maps as follows. The cokernel $B_i(\mathbf{G}) := \delta_{i+1}(C_{i+1}(\mathbf{G})) \leq C_i(\mathbf{G})$ of δ_{i+1} is the ith *boundary group*; its elements are i-boundaries. The kernel $Z_i(\mathbf{G}) = \ker(\delta_i) \leq C_i(\mathbf{G})$ is the ith *cycle group*; its elements are *i-cycles*. It is easy to check that $\delta_i \circ \delta_{i+1} = 0$ and so $B_i(\mathbf{G}) \leq Z_i(\mathbf{G})$. The ith *homology group* is $H_i(\mathbf{G}) = Z_i(\mathbf{G})/B_i(\mathbf{G})$; its elements are classes of *homologically equivalent* cycles, though we often refer to these classes simply as cycles.

We start with some standard easy calculations.

Example 3.1. For any graph G, $H_0(\mathbf{G}) = \mathbb{Z}$. Indeed, $C_0(\mathbf{G}) = \langle [v] \mid v \in V(G) \rangle$ and as δ_{-1} is the 0 map, $Z_0(\mathbf{G}) = \ker \delta_{-1} = C_0(\mathbf{G})$. On the other hand, $C_1(\mathbf{G}) = \langle [u, v] \mid uv \in E(G) \rangle$, so $B_0(\mathbf{G}) = \delta_1(C_1) = \langle [u] - [v] \mid uv \in E(G) \rangle$. Thus $H_0(\mathbf{G}) = Z_0(\mathbf{G})/B_0(\mathbf{G}) = \mathbb{Z}$. △

⟨example:basprop⟩ **Example 3.2.** The following are all true for any triangle-free G.

(1) $C_2(\mathbf{G})$ is the free group generated by the set
$$\{[v,v,v] \mid v \in V(G)\} \cup \{[u,u,v],[u,v,u],[v,u,u] \mid uv \in E(G)\}$$
(2) As $\delta([v,v,v]) = [v,v]$, and, for example, $\delta([u,v,u]) = [v,u] - [u,u] + [u,v]$, one finds that $B_1(\mathbf{G}) = \delta_2(C_2)$ is free generated by
$$\{[v,v] \mid v \in V(G)\} \cup \{[u,v] - [v,u] \mid uv \in E(G)\}$$
(3) $Z_1(\mathbf{G})$ is free generated by cycles (including loops and 2-cycles) in G.
(4) $H_1(\mathbf{G}) = Z_1(\mathbf{G})/B_1(\mathbf{G})$ is free generated by the undirected edge-sets of cycles (not-including loops and 2-cycles) in G.
(5) $H_n(\mathbf{G})$ is *trivial*; that is, has only the 0 element; for $n \geq 2$.

△

In light of (3), we will often talk of a cycle $(x_0, x_1, \ldots, x_\ell, x_0)$ of G as being in $Z_1(\mathbf{G})$. This really means the 1-cycle
$$[x_0, x_1] + [x_1, x_2] + \cdots + [x_\ell, x_0].$$

Example 3.3. If G is not triangle free, then a triangle such as (u,v,w,u), which lives in $Z_1(\mathbf{G})$ as $\sigma = [u,v] + [v,w] + [w,u]$ is a boundary in $B_1(\mathbf{G})$:
$$\delta_2([u,v,w]) = [v,w] - [u,w] + [u,v] = [u,v] + [v,w] + [w,u],$$
so $[\sigma] = 0$ in $H_1(\mathbf{G})$. Thus we still have that $H_1(\mathbf{G})$ is generated by undirected edges-sets of cycles in G, but it may no longer be free, and there can be more complicated boundaries as well. The groups H_n for $n \geq 2$ are also not generally trivial.

△

The group $Z_1(\mathbf{G})$ of 1-cycles is what is often called the integral cycle space of G; as opposed to the cycle space, which we get with the same development using $\mathbb{Z}_2$ in place of $\mathbb{Z}$. (Often though $C_2(\mathbf{G})$ is taken to be empty, so $Z_1(\mathbf{G})$ is called $H_1(\mathbf{G})$.) A cycle C in G is *trivial* if $[C] = 0$ in $H_1(\mathbf{G})$, where 0 represents the class containing the empty cycle. If $H_1(\mathbf{G}) = \{0\}$ then we say G has *trivial homology*. An NT-*basis* of G is a basis of $H_1(G)$; NT refers to the fact that it contains No Triangles. With triangles of G it makes up a basis of the integral cycle space of G.

⟨fact:preshom⟩ **Fact 3.4.** *Let $\phi : P_\ell \times C \to H$ be a homomorphism. For all $i, j \in \{0, 1, \ldots, \ell\}$ we have $[\phi(i \times C)] = [\phi(j \times C)]$ in $H_1(H)$.*

Proof. Observe that the boundary $\delta_2(c_i)$ of the 2-chain $c_i := [(i,i),(i,i+1),(i+1,i)] + [(i+1,i),(i,i+1),(i+1,i+1)]$ is the square
$$[(i,i),(i,i+1)] + [(i,i+1),(i+1,i+1)] + [(i+1,i+1),(i+1,i)] + [(i+1,i),(i,i)].$$
So the boundary of the 2-chain $\sum_{i=0}^{\ell} c_i$ is $(0 \times C) - (1 \times C)$, and so its image $\phi(0 \times C) - \phi(1 \times C)$ is also a boundary. Thus $[\phi(0 \times C)] = [\phi(1 \times C)]$ in $H_1(H)$ and the fact follows by induction. □

4. Reduction to the problem NonFlat($\mathbf{H}$)

?⟨sect:non-flat⟩? In this section we reduce our main result to a problem called *non-flat H-colouring*.

?⟨def:flat⟩? **Definition 4.1.** A homomorphism $\phi : G \to H$ is *flat* if for all cycles C in G, $[\phi(C)] = 0$ in $H_1(H)$. Equivalently ϕ is flat if $[\phi(C)] = 0$ is contractible for all cycles C in an NT-basis of G.

Consider the following problem.

Problem: NonFlat(H)
Instance: A graph G.
Question: Is there a non-flat homomorphism of G to H.

We will prove the following.

⟨thm:NF⟩ **Theorem 4.2.** *Let H be a reflexive graph such that $H_1(H)$ is free and non-trivial. The problem* NonFlat(H) *is* NP*-hard.*

The proof of this takes up most of the paper. For the rest of this section, we show that it implies Theorem 1.2. For this we need a result from [**?**]. We defined homotopy classes of cycles, similar to homology classes, and called a cycle contractible if it was in the same class as the constant

?⟨def:moves⟩? **Definition 4.3.** Let $\Pi(H; x_0)$ be the graph whose vertices are cycles in H starting at x_0, and in which a cycle Y is adjacent to a cycle $X = (x_0, \ldots, x_\ell, x_0)$ if either of the following are true:

(P1) $Y = (x_0, x_1, \ldots, x_{i-1}, x_i, x_i, x_{i+1}, \ldots, x_\ell, x_0)$ for any i, or
(P2) $Y = (x_0, x_1, \ldots, x_{i-1}, x_i', x_{i+1}, \ldots, x_\ell, x_0)$ for some $i \notin \{0, \ell\}$, where $x_{i-1} = x_{i+1}$.

Let $\Pi(H)$ be the graph we get from the disjoint union, over all $x_0 \in V(H)$, of $\Pi(H; x_0)$ by adding an edge between walks y and $x = (x_0, x_1, \ldots, x_{\ell-1}, x_0)$ if

(P3) $y = (x_1, \ldots, x_{\ell-1})$ and $x_1 = x_{\ell-1}$.

Observe that if H is connected all the constant cycles, those in which all vertices are the same, are in the same component of $\Pi(H)$. A cycle is *contractible* if it is in the same component of $\Pi(H)$ as the constant cycles.

The following is from [**?**].

⟨cor:LNS⟩ **Corollary 4.4.** *A homomorphism ϕ in* $\mathbf{Hom}(G, H)$ *is in the constant component of* $\mathbf{Hom}(G, H)$ *if for every cycle C in G, $\phi(C)$ is contractible in $\Pi(H)$.*

Now observe. the following analog of a well known idea of topology.

⟨lem:homhom⟩ **Lemma 4.5.** *Let H be a triangle-free reflexive graph. If $[\phi(C)]$ is contracible in $H_1(H)$, then $\phi(C)$ is contractible in $\Pi(H)$.*

Proof. Our proof is by induction on the length of $\phi(C)$, it is clearly true of cycles of length 1. Let $\phi(C) = (x_0, \ldots, x_\ell, x_0)$, for $\ell \geq 1$. As a cycle in $Z_1(G)$ we write this as

$$[x_0, x_1] + [x_1, x_2] + \cdots + [x_\ell, x_0].$$

If the negation of some element in the chain appears later in the chain: $[x_j, x_{j+1}] = x_{i+1}, x_i$ for some $i < j$, then these disappear in $H_1(G)$, and we get in $H_1(G)$ that $[\phi(C)] = [C_1] + [C_2]$ where

$$C_1 = [x_{i+1}, x_{i+2}] + \cdots + [x_{j-1}, x_j] \text{ and } C_2 = [x_{j+1}, x_{j+1}] + \cdots + [x_{i-1}, x_i]$$

with indices modulo ℓ (one of these might be empty). The result therefore follows by induction.

So we may assume that the 1-chain $\phi(C) = [x_0, x_1] + [x_1, x_2] + \cdots + [x_\ell, x_0]$ consists of non-negated summands. But as $Z_1(G)$ is free and $B_1(G)$ is as given in Example 3.2(2) the fact that $\phi(C) = 0$ means that all summands in this chain are $[x_0, x_0]$, and so $\phi(C)$ is constant, and so contracible in $\Pi(H)$. □

Thus from Corollary 4.4 we get the following.

Fact 4.6. *If $\phi : G \to H$ is flat in $H_1(\mathbf{G})$ then it is in the constant component of $\mathbf{Hom}(G, H)$. If all homomorphisms $\phi : G \to H$ are flat, then $\mathbf{Hom}(G, H)$ is connected.*

⟨lem:3toNF⟩ **Lemma 4.7.** *If* NonFlat(H) *is* NP*-complete for a triangle-free reflexive cycle H, then* Mix(H) *is* coNP*-complete.*

Proof. We observe that the negation of NonFlat(H) can be solved by Mix(H). Indeed let G be an instance of NonFlat(H). If it is a YES instance, then there is a non-flat homomorphism $\phi : G \to H$. Thus for some cycle B of G, $[\phi(B)] \neq 0$. By Lemma 4.5 we have then that ϕ in non-contracible in $\Pi(H)$, so G is NO instance of Mix(H).

On the other hand, if G is a NO instance of NonFlat(H) the all homomorphisms $\phi : G \to H$ are flat, and so by Lemma 4.5 and Corollary 4.4 reconfigure to the constant homomorphisms. Thus G is a YES instance of Mix(H). □

Observe now that Theorem 1.2 follows from Theorem 4.2.

Proof of Theorem 1.2. Let H be a triangle-free reflexive graph containing a cycle. By Example 3.2(4) $H_1(\mathbf{H})$ is free, and as it contains a cycle, it is non-trivial, so we may apply Theorem 4.2 to get that NonFlat(H) is NP-complete. Lemma 4.7 then tells us that Mix(H) is coNP-complete. So Theorem 1.2 is proved. □

For the rest of the paper, we therefore concentrate on proving Theorem 4.2.

5. The Homology Sum gadget S_s

⟨sect:windsum⟩ Let H_* be a connected reflexive graph with $H_1(H_*) \neq 0$, and let C_* be a shortest cycle in H_* with $[C_*] \neq 0$ in $H_1(H_*)$. Let $g \geq 4$ be the girth of C_*. Let B_* be the symmetric reflexive cycle of girth $s \cdot g$.

Observing that the vertex set of B_* is $V(B_*) = \{ig + j \mid i \in [s], j \in [g]\}$, define the following maps that are clearly in $\mathrm{Hom}(B_*, C_*)$. Let $\beta : B_* \to C_*$ be defined by

$$\beta(ig + j) = i,$$

and for $k \in [s]$ let $\alpha_k : B_* \to C_*$ be defined by

$$\alpha_k(ig + j) = \begin{cases} j & \text{if } k = i \\ 0 & \text{if otherwise.} \end{cases}$$

So β maps the first s vertices of B_* to 0, the next s vertices to 1, etc., while α_k maps all vertices to 0 except the kth run of g vertices which it maps around C_*. Observe, in the case of $(s, g) = (3, 4)$ for example, there that are paths in $Hom(B_*, C_*)$ of the form

$$000111222333 \sim 011111222333 \sim 012222222333 \sim 012333333333 \sim 012300000000$$

from β to α_0, and

$$012300000000 \sim 001230000000 \sim 000123000000 \sim \cdots \sim 000001230000$$

from α_0 to α_1, in which the first vertex is always 0. From this it is not too hard to see that the following is true.

⟨fact:lexists⟩ **Fact 5.1.** *There is an integer $\ell = \ell_s \geq 2$ such that for each $i \in [s]$ there is a homomorphism $\Phi : P_\ell \times B_* \to C_*$ that*

(1) $\Phi_0 = \alpha_i$*, and*
(2) $\Phi_\ell = \beta$*, and*
(3) Φ *is constant* 0 *on* $T := P_\ell \times 0$*.*

With this ℓ we are now ready to construct S_s, as shown in Figure **??**.

⟨const:sum⟩ **Construction 5.2.** *Let* S_s *be constructed from* $\mathcal{P} = P_\ell \times B_*$*, where* $\ell = \ell_s$ *, as follows.*

(1) In the last slice $\ell \times B_*$ *of* $\mathcal{P}$*, for each* $j \in [g]$ *identify the vertices in the set*
$$\{(\ell, i + js) \mid i \in [s].\}$$
Refer to the resulting copy of C_* *as* Z*.*

(2) In the first slice $0 \times B_*$ *of* $\mathcal{P}$*, identify the vertices in the set*
$$\{(0, ig) \mid i \in [s].\}$$
This creates s *copies of* C_* *sharing vertex. Refer to these as* $A'_1, \ldots, A'_s$*.*

(3) For each $i \in [s]$*, take a new copy of* $P_1 \times C_*$*, identify one slice of it with* A'_i *and call the other slice* A_i*.*

Let $S_s^{(i)}$ be the graph constructed by step (i) of the construction. Observe that the identifications in steps (1) and (2), (and even the injection in step (3)), can be viewed as homomorphisms $q_1 : \mathcal{P} \to S_s^{(1)}$ and $q_2 : S_s^{(1)} \to S_s^{(2)}$. Notationally we should use names such as $q_2 \circ q_1((\ell, 0))$ for vertices of $S'_s = S_s^{(3)}$ but the abuse of calling this $(\ell, 0)$ is convenient and does not cause problems.

This allows us the following definition that we will need later.

Definition 5.3. A cycle in S_s is a *gadget-cycle* if it is $(q_2 \circ q_1)(B)$ for some slice $B = i \times B_*$ of $\mathcal{P}$ or is A_i for some i. The cycles A_i and Z are *end gadget-cycles.*

With this observation we prove the following desired properties of S_s.

⟨gad:sum⟩ **Lemma 5.4.** *The graph* $S = S_s$ *contains labelled copies* $Z, A_0, \ldots, A_{s-2}$ *and* A_{s-1} *of* C_**, which are pairwise distance at least* 2 *apart; and a tree* $T = T(S)$ *that contains the copy of* 0 *in each of these copies of* C_**. Further, for any* H_* *whose smallest homologically non-trivial cycle is* C_* *the following are true.*

(1) For any homomorphism $\Gamma : S \to H_*$*, we have* $\Gamma(Z)$ *is either a copy of* C_* *or homologically trivial and,*
$$[\Gamma(Z)] = \sum_{i \in [s]} [\Gamma(A_i)].$$

(2) For each $i \in [s]$ *there is a homomorphism* $\Gamma_i : S \to C_*$ *such that* $[\Gamma_i(A_i)] = [C_*]$ *and such that* Γ_i *is* 0 *on* T *and* A_j *for* $j \neq i$*.*

(3) (I want to put this first, but have to worry about references.) For any hom $\Gamma : S \to H_*$ *with* $[\Gamma(Z)] = 0$*, all gadget cycles* C *except possibly the* A_i *have* $[\phi(C)] = 0$*.*

Proof. The statements before the 'Further' are just trivial observations. For part (1), let Γ be any homomorphsim of S to H_*. As Z has the same girth g as the minimum cycles C_* of H_*, $\Gamma(Z) \in \{[C_*], 0\}$ for some copy of C_* in H_* is clear. To see the rest of part (1) observe that the homomorphism $\Gamma' = \Gamma \circ (q_2 \circ q_1) : \mathcal{P} \to S_s^{(2)} \to C_*$ satisfies
$$[\Gamma'(\ell \times C_*)] = [\Gamma(Z)] \text{ and } [\Gamma'(0 \times C_*)] = \sum [\Gamma(A'_i)].$$

As the left sides of these equalities are equal by Fact **??**, so are the right sides. Using Fact **??** on the path of cycles connecting A_i and A'_i we get (1).

Now let T be the tree in S_s induced by the vertices $(i, 0)$ in $S_s^{(2)}$ and the vertex 0 in A_i for each i. To get property (2), let $\Gamma'_i = (q_2 \circ q_1) \circ \Phi_i$ where Φ_i is from Fact 5.1, and extend Γ'_i to $\Gamma_i : S_s \to C$ by letting it act on A_i as is does on A'_i.

TO GET PROPERTY (3) ...

□

6. The preliminary construction

⟨sect:stage1⟩ Having built our gadget S_s, our main construction, that of G^*, will be completed in two stages. In this section, we do the first stage, constructing, for every (irreflexive) instance G of 3-colouring, a graph $G^{(a)}$ containing a designated copy Z_* of C_*, which satisfies the properties listed in Lemma 6.4. This construction is outlined in Section 2.

⟨const:G*⟩ **Construction 6.1.** *For an irreflexive graph G construct $G^{(a)}$ as follows.*

(1) For every vertex v of G let $G^{(a)}$ contain three copies A_0^v, A_1^v and A_2^v of C_*. Let $G^{(a)}$ also contain a copy Z_* of C_*. Let all of these be disjoint.

(2) For every vertex v of G create a new copy S^v of the sum gadget S_3; identify the copy of Z in S^v with Z_*, and for each $i \in [3]$ identify the copy of A_i in S^v with A_i^v .

(3) For every vertex v of G and every pair of distinct indices $i < j \in [3]$, create a new copy $S_{i,j}^v$ of S_2; identify its copy of A_0 with A_i^v and its copy of A_1 with A_j^v. However, in identifying this copy of A_1 with A_j^v identify the copy of the vertex i in A_1 with the vertex $g - i$ in A_j^v.

(4) For each edge uv in G and each $i \in [3]$, create a new copy S_i^{uv} of S_2; identify its copy of A_0 with A_i^u and its copy of A_1 with A_i^v.

Let $G^{(a.i)}$ refer to the graph constructed in step (i) of the construction.

For each sum gadget S used in the construction, Lemma 5.4 gives us a subgraph $T(S)$ containing the vertex 0 of every gadget-cycle. Let $T_* \leq G^{(a)}$ be the union of all these $T(S)$.

First we observe some easy properties of the construction, then prove the main properties.

Definition 6.2. A cycle of $G^{(a)}$ is a *gadget-cycle* if it is a gadget-cycle in one of the gadgets S_s used in constructing $G^{(a)}$.

Fact 6.3. *Where $G^{(a)}$ is constructed from a graph G by Construction 6.1, the following are true.*

(1) The subgraph $T_* \leq G^{(a)}$ is connected.

(2) The order $|V(G^{(a)})|$ is polynomial in $|V(G)|$.

Proof. To see (1) observe that $G^{(a)}$ is the union of copies of S_3 and S_2 that are attached by identifying end gadget-cycles. The vertices 0 in these, which are in T_*, are always identified.

For (2) we observe that $G^{(a)}$ consists of $|V(G)| = O(n)$ copies of S_3 and $3(|V(G)|+|E(G)|) = O(n^2)$ copies of S_2; these have constant order. □

We now prove the main properties of the construction.

⟨lem:reduct1⟩ **Lemma 6.4.** *There is a homomorphsim $\phi : G \to K_3$ if and only if there is a homomorphism $\phi^* : G^{(a)} \to H_*$ such that $[\phi^*(Z_*)] = [C]$ for some copy C of C_* in H_*. Moreover, if there is such a homomorphism $\phi^* : G^{(a)} \to H_*$, we may assume that it is constant 0 on the subgraph T_* of $G^{(a)}$.*

Before we prove Lemma 6.4, we prove one fact, that we will use again later. This was the point of step (2) of Construction 6.1.

⟨claim:exactly1⟩ **Claim 6.5.** *For any homomorphism $\phi^* : G^{(a)} \to H_*$ and any $v \in V(G)$, either*

(1) $[\phi^*(Z)] = 0$ and $[\phi^*(A_i^v)] = 0$ for all $i \in [3]$, or
(2) $[\phi^*(Z)] \neq 0$ and exactly one of the $[\phi^*(A_i^v)]$ are equal to this, the other two are 0.

Proof. By Fact **??**, $H_1(\mathbf{H}_*)$ is a free group. As Z and A_i^v have the girth of the minimal cycle of H_*, their images under ϕ^* are either 0 or generators of $H_1(\mathbf{H}_*)$. The paths-of-cycles introduced in step (2) of Construction 6.1 preclude the possibility that two of the $[\phi^*(A_i^v)]$ have opposite signs, so item (1) of Lemma 5.4 yields the claim. □

We are now ready to prove Lemma 6.4.

Proof. One the one hand, assume that there is a homomorphism $\phi^* : G^{(a)} \to H_*$ with $0 \neq [\phi^*(Z_*)] =: [C]$. To define $\phi : G \to K_3$, consider some $v \in V(G)$. As Z_* is identified with the copy of Z in $S^v \cong S_3$, we have by (1) of Lemma 5.4 and Claim 6.5, that $[A_i^v] = [C]$ for **exactly** one $i \in [3]$. Let $\phi(v)$ be this i. This defines a mapping $\phi : V(G) \to V(K_3)$. Now, for any edge uv of G we have $\phi(u) \neq \phi(v)$, as if $\phi(u) = i = \phi(v)$ then ϕ^* restricted to $S_i^{uv} \cong S_2$ has $[\phi^*(A_0)] = [\phi^*(A_1)] = [C_*]$, which violates (1) of Lemma 5.4. Thus $\phi : G \to K_3$ is a homomorphism.

Assume, on the other hand, that there is a homomorphism $\phi : G \to K_3$. We define $\phi^* : G^{(a.2)} \to H_*$ by choosing a particular copy C of C_* in H_* and defining $\phi^* : G^{(a.2)} \to C$ as follows. For each v let ϕ^* restricted to S^v be $\Gamma_{\phi(v)}$ from Lemma 5.4. These all agree on the copies of Z, so this is well defined, and has $[\phi^*(Z_*)] = [C]$ as needed. Further, if $\phi(v) = i$ then ϕ^* is constant 0 on A_j^v for $j \neq i$ and maps $A_i^v \cong C_*$ identically to C.

Now we extend ϕ^* to a homomorphism $G^{(a.3)} \to C$. For each vertex u of G, let the copy $S_{i,j}^u$, for $i < j$, be coloured with the colouring Γ_0 if $\phi(v) = i$, with colouring Γ_1 if $\phi(v) = j$, and with the constant 0 colouring otherwise. This agrees with ϕ^* defined on $G^{(a.2)}$ on the cycles A_i^v, so is well defined.

Finally, we extend ϕ^* to a homomorphism $G^{(a)} \to C$. For each edge uv of G and $i \in [3]$, at most one of $\phi(u)$ and $\phi(v)$ is i. Let ϕ^* be Γ_0 (from Lemma 5.4) on S_i^{uv} if $\phi(u) = i$, be Γ_1 if $\phi(v) = i$, or be constant 0 if neither of these holds. This agrees with the definition of ϕ^* on $G^{(a.2)}$, so is well defined. □

Before we move onto the second stage of the construction, which will allow us to prove the required converse to Lemma 6.4, we observe there is already a partial converse.

⟨lem:partconv⟩ **Lemma 6.6.** *For a homomorphism $\phi\colon G^{(a)} \to H_*$ such that $[\phi(Z_*)] = 0$, we have that $[\phi(C)] = 0$ for every gadget-cycle C of $G^{(a)}$.*

Proof. This is immediate from property (3) of Lemma 5.4 and Claim 6.5.

MOVE THE PROOF TO LEMMA 5.4:

Let $\phi : G^{(a)} \to H_*$ be a homomorphism with $[\phi(Z_*)] = 0$. By Fact **??** we get, for a copy of the gadget S_s that if $[\phi(Z)] = 0$ f if the copy of Z gets wind 0, then all gadget-cycles get wind 0, except possibly the copies of A_i and A'_i for all i, thought the sums $\sum \mathrm{wind}_\phi(A_i)$ and $\sum \mathrm{wind}_\phi(A'_i)$ are 0.

Consider the copies S^v of S_v in $G^{(a.1)}$. By the above observation, and Claim 6.5, all of their gadget-cycles get wind 0.

All other copies of S_s used in $G^{(a)}$ have all but one of their end gadget-cycles identified with one of the gadget-cycles already shown to have wind 0; so by Lemma 5.4(1), and Fact **??** all gadget-cycles have wind 0. □

7. Computing the NT-basis of $G^{(a)}$

⟨sect:NTbasis⟩ Recall that T_* is the subgraph of $G^{(a)}$ consisting of the special subgraph $T(S)$ of each of the sum gadgets S used in its construction. In this section, we prove the following.

⟨thm:topo⟩ **Theorem 7.1.** *$G^{(a)}$ has a* NT*-basis consisting only of gadget-cycles and the cycles in an NT-basis of T_*.*

The graph $G^{(a)}$ is constructed from many copies of the paths-of-cycles by taking quotients (which includes contracting edges), and gluing things together. Homology provides us the tools of excision and Mayer-Vietoris sequences to compute homology groups under these operations. Working through the steps of Constructions 5.2 and 6.1 with these computational tools we will find NT-bases of S_s and then $G^{(a)}$. First though, as a path-of-cycles is a product, we compute its homology via a Künneth formula.

7.1. **The NT-basis of a path-of-cycles.** Our notation $\mathbf{P}_\ell \times \mathbf{C}_*$ is ambiguous as the product space of the clique complexes $\mathbf{P}_\ell$ and $\mathbf{C}_*$ is not the same as clique-complex of the product $P_\ell \times C_*$. Indeed the product $P_\ell \times C_*$ is shown on the left of Figure **??** as a tiling of a cylinder with copies of K_4, while the product space $\mathbf{P}_\ell \times \mathbf{C}_*$ (see [**?**] pp.277-278), shown on the right, is a triangulation of a cylinder. It is not too hard to see that the triangulation embeds in the K_4-tiling, and that the K_4-tiling deformation retracts to the triangulation. So the two simplicial complexes have isomorphic homology groups. For the following calculation, therefore, which says that any one gadget-cycle in a path-of-cycles can be taken as an NT-basis for the path-of-cycles, we may assume that these two products coincide.

⟨lem:hcylinder⟩ **Lemma 7.2.** *For any integer $\ell \geq 1$ and $d \geq 4$, and any $i \in [\ell]$,*

$$H_1(\mathbf{P}_\ell \times \mathbf{C}_*) = \langle i \times C_* \rangle \cong \mathbb{Z}.$$

Proof. The Künneth formula (see Theorem 3B.7 of [**?**] gives us that

$$H_k(\mathbf{P}_\ell \times \mathbf{C}_*) = \bigoplus_{i+j=k} H_i(\mathbf{P}_\ell) \otimes_{\mathbb{Z}} H_j(\mathbf{C}_*)$$

where $\oplus$ is the direct product of groups and $\otimes_{\mathbb{Z}}$ is the tensor product over $\mathbb{Z}$. Thus we get

$$\begin{aligned} H_1(\mathbf{P}_\ell \times \mathbf{C}_*) &= (H_1(\mathbf{P}_\ell) \otimes_{\mathbb{Z}} H_0(\mathbf{C}_*)) \oplus (H_0(\mathbf{P}_\ell) \otimes_{\mathbb{Z}} H_1(\mathbf{C}_*)) \\ &= (0 \otimes_{\mathbb{Z}} \mathbb{Z}) \oplus (\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}) \\ &= 0 \oplus \mathbb{Z} = \mathbb{Z}, \end{aligned}$$

as needed. That the generator can be taken as an $i \times C_*$ seems clear. □

7.2. **Excision and Mayer-Vietoris Sequences.** Our main tools are excision for dealing with quotients, and Mayer-Vietoris sequences for patching together gadgets. For these we will use the *reduced homology groups* $\tilde{H}_i(\mathbf{G})$, which are defined on page 210 of [**?**]. When $i \geq 1$ we have $\tilde{H}_i(\mathbf{G}) = H_i(\mathbf{G})$, but for $i = 0$ we have $\tilde{H}_0(\mathbf{G}) \oplus \mathbb{Z} \cong H_0(\mathbf{G})$. So whereas the dimension of $H_0(\mathbf{G})$ is the number of components of G, the dimension of $\tilde{H}_0(\mathbf{G})$ is one less than the number of components. Generally we use connected G so $\tilde{H}_0(\mathbf{G})$ is usually 0.

Recall also that a sequence of groups

$$\cdots \to M_{i+2} \xrightarrow{\alpha_{i+2}} M_{i+1} \xrightarrow{\alpha_{i+1}} M_i \xrightarrow{\alpha_i} M_{i-1} \to \dots$$

is exact if for each i, $\alpha_i \circ \alpha_{i+1} = 0$, and that if a we have a short exact sequence of groups

$$0 \to A \to B \to C \to 0,$$

where C is free, then the sequence splits (see [**?**] p. 148), which means that $B = A \oplus C$.

The following is a special case of the process known as excision. It tells us that when contracting an edge not in a C_4, our NT-basis is unchanged.

⟨lem:hexcision⟩ **Lemma 7.3.** *Let uv be an edge of G that is in no induced copy of C_4, and let $G' = G/uv$ be the graph we get by contracting the edge uv. Then $[C] \mapsto [C/uv]$ is an isomorphism $H_1(\mathbf{G}) \cong H_1(\mathbf{G}')$.*

Proof. As uv is in no C_4, contracting it makes no new K_3, so no new 2-simplices. Thus, as simplicial complexes, $\mathbf{G}'$ is exactly the quotient $\mathbf{G}/\mathbf{E}$ where $\mathbf{E}$ is the simplicial comlpex of the edge uv.

Take an epsilon neighbourhood in $\mathbf{G}$ around $\mathbf{E}$. It consists of $\mathbf{E}$ and little nubs of edges and some frills of triangles. These all deformation retract to $\mathbf{E}$ so we can apply excision (Thm 2.13 of [**?**]) to get that

$$H_1(\mathbf{E}) \to H_1(\mathbf{G}) \xrightarrow{j_*} H_1(\mathbf{G}/\mathbf{E}) \to \tilde{H}_0(\mathbf{E})$$

is exact, where j_* is induced by the quotient $j : \mathbf{G} \to \mathbf{G}'$. As $\tilde{H}_0(\mathbf{E}) = 0$, and $H_1(\mathbf{E}) = 0$, j_* is an isomorphism $H_1(\mathbf{G}) \cong H_1(\mathbf{G}')$ as needed. □

Finally, we recall the basic idea of Mayer-Vietoris sequences, and use it for several lemmas. Given spaces $\mathbf{A}$ and $\mathbf{B}$ that intersect in $\mathbf{C} = \mathbf{A} \cup \mathbf{B}$, we can often compute the homology of $\mathbf{X} = \mathbf{A} \cup \mathbf{B}$ using the fact that the following sequence is exact.

$$\dots \xrightarrow{\delta_{n+1}} H_n(\mathbf{C}) \xrightarrow{\Phi_n} H_n(\mathbf{A}) \oplus H_n(\mathbf{B}) \xrightarrow{\Psi_n} H_n(\mathbf{X}) \xrightarrow{\delta_n} H_{n-1}(\mathbf{C}) \xrightarrow{\Phi_{n-1}} \tag{1}$$

eq:MV

$$\dots \xrightarrow{\Psi_1} H_1(\mathbf{X}) \xrightarrow{\delta_1} \tilde{H}_0(\mathbf{C}) \xrightarrow{\Phi_0} \tilde{H}_0(\mathbf{A}) \oplus \tilde{H}_n(\mathbf{B}) \xrightarrow{\Psi_0} \tilde{H}_0(\mathbf{X}) \longrightarrow 0$$

where $\Psi(\alpha, \beta) = \alpha - \beta$. See [**?**] p.149; on p. 150 it is stated that it holds, as given above, for reduced homology.

This lemma, which we use a couple of times, is nearly obvious. But being the most simple use of a Mayer-Vietoris sequence, is a useful example.

⟨lem:hMV0⟩ **Lemma 7.4.** *Let G be a connected graph and a and b be vertices of G at distance at least 4 apart. Where $G+E$ is the graph we get from G by adding the edge $E = ab$, $H_1(\mathbf{G}+\mathbf{E}) = H_1(\mathbf{G}) \oplus \mathbb{Z}$. Moreover one gets a basis of $H_1(\mathbf{G}+\mathbf{E})$ from one of $H_1(\mathbf{G})$ by adding any cycle C containing one copy of the edge E.*

Proof. As a and b are distance at least 4 apart, we make no new triangles by adding E. So by (1) the following is exact.

$$\to H_1(\{\mathbf{a},\mathbf{b}\}) \to H_1(\mathbf{G}) \oplus H_1(\mathbf{E}) \xrightarrow{\Psi} H_1(\mathbf{G}+\mathbf{E}) \to \tilde{H}_0(\{\mathbf{a},\mathbf{b}\}) \to \tilde{H}_0(\mathbf{G}) \oplus \tilde{H}_0(\mathbf{E}) \to$$

Using $H_1(\{\mathbf{a},\mathbf{b}\}) = 0$, $H_1(\mathbf{E}) = 0$ and $\tilde{H}_0(\mathbf{G}) \oplus \tilde{H}_0(\mathbf{E}) = 0 \oplus 0 = 0$ we get a short exact sequence

$$0 \to H_1(\mathbf{G}) \oplus 0 \xrightarrow{\Psi} H_1(\mathbf{G}+\mathbf{E}) \xrightarrow{\delta} \tilde{H}_0(\{\mathbf{a},\mathbf{b}\}) \to 0.$$

As $\tilde{H}_0(\{\mathbf{a},\mathbf{b}\}) = \mathbb{Z}$ is free, this splits to give $H_1(\mathbf{G}+\mathbf{E}) = H_1(\mathbf{G}) \oplus \tilde{H}_0(\{\mathbf{a},\mathbf{b}\}) = H_1(\mathbf{G}) \oplus \mathbb{Z}$ as needed.

The statement about the basis comes as follows: the injection Ψ injects $H_1(\mathbf{G})$ as a subgroup of $H_1(\mathbf{G}+\mathbf{E})$. By exactness, this is exactly the kernel of δ, so one gets a final basis element of $H_1(\mathbf{G}+\mathbf{E})$ from any cycle that δ maps onto the generator $[b-a]$ of $\tilde{H}_0(a,b)$. This is any cycle made up of a walk in G with boundary $[b-a]$ and a walk in E with boundary $[a-b]$; so any cycle C in G that contains E once. □

The following lemma uses the above, with excision to say that we 'pinch' a long basis cycle into 2 cycles, they both become basis cycles, replacing the original cycle in the basis. This will be used in Step (2) of the construction of the sum gadget.

⟨lem:hMV1⟩ **Lemma 7.5.** *Let G be a graph, C be a non-contractible cycle in $H_1(\mathbf{G})$, and a and b be vertices of C at distance at least 4 apart. Where G' is the graph we get from G by identifying a and b, $H_1(\mathbf{G}') = H_1(\mathbf{G}) \oplus \mathbb{Z}$. Moreover, one gets a basis of $H_1(\mathbf{G}')$ by taking a basis of $H_1(\mathbf{G})$, and doing either of:*

- *adding one of the new cycles made from C by identifying a and b, or*
- *replacing $[C]$ with new cycles we get from C by the identification.*

Proof. Add an edge E between a and b. By Lemma 7.3 and Lemma 7.4 we get

$$H_1(\mathbf{G}') = H_1(\mathbf{G}+\mathbf{E}) = H_1(\mathbf{G}) \oplus \mathbb{Z},$$

as needed.

To get a basis of $H_1(\mathbf{G}+\mathbf{E})$ from that of $H_1(|DG)$ we add any cycle containing E once. So we us the cycle $C_1 = P_1 + E$ where P_1 is a path in C from b to a. Moreover, where $C = P_2 + P_1$, we have $[C] = [C_2] + [C_1]$, so we get another basis of $H_1(\mathbf{G}+\mathbf{E})$ by removing $[C]$ and adding $[C_2]$.

Under the the contraction of E to $\mathbf{G}'$, C_1 and C_2 become the cycles we get from C by identifying a and b, as needed. □

The following says that if we identify end gadget-cycles of disjoint gadgets then, assuming that the identified gadget-cycle is in an NT-basis of both gadgets, our NT-basis in the new graph is the union of the NT-bases of the two gadgets. (The the identified gadget-cycle only appears once in the new basis.)

⟨lem:hjoin1⟩ **Lemma 7.6.** *Let F and H be disjoint connected graphs and let C_F and C_H be isomorphic non-contractible 1-cycles of F and H respectively. Let G be the graph we get from F and H by identifying C_F and C_H. Then*

$$H_1(\mathbf{G}) = (H_1(\mathbf{F}) \oplus H_1(\mathbf{H}))/H_1(\mathbf{C_F}).$$

(This is '$=$', not just '$\cong$', so one gets an NT*-basis of G from the union of bases of F and H, by removing the repeated basis element $[C_F]$.)*

Proof. By (1), the sequence

$$\cdots \to H_2(\mathbf{G}) \xrightarrow{\delta} H_1(\mathbf{C_F}) \xrightarrow{\Phi} H_1(\mathbf{F}) \oplus H_1(\mathbf{H}) \xrightarrow{\Psi} H_1(\mathbf{G}) \to \tilde{H}_0(\mathbf{C_F}) \to \dots$$

where $\Phi(\alpha) = (\alpha, -\alpha)$, is exact. As $\tilde{H}_0(\mathbf{C_F}) = \emptyset$, the map Ψ, which is isomorphic on both $H_1(\mathbf{F})$ and $H_1(\mathbf{H})$, is surjective. So elements of $H_1(\mathbf{F})$ and $H_1(\mathbf{H})$ generate $H_1(\mathbf{G})$.

As C_F is non-zero in $H_1(\mathbf{G})$ it is not the boundary of anything in G, so δ maps everything to 0. Thus by exactness, Φ is injective, and the kernal of Ψ is the element $(C_F, -C_F) = (C_F, -C_H)$, identifying C_F and C_H. So $H_1(\mathbf{G}) = (H_1(\mathbf{F}) \oplus H_1(\mathbf{H}))/H_1(\mathbf{C_F})$, as needed.

□

The following two lemmas deal with the situation when we identify two gadget-cycles of a connected graph. In doing so, one of the identified gadget-cycles, both of which can be assume to be basis cycles, becomes dependent on the other, so falls out of the basis; at the same time, we create a new basis cycle that we can take to be in T_*.

The lemmas work much like Lemmas 7.4 and reflem:hMV1 worked to identify two vertices. Lemma 7.7 adds a new path-of-cycles between the cycles we want to identify, and then Lemma 7.8 uses excision to 'contract' this path.

We say cycles are *independent* if they are independent in $H_1(\mathbf{G})$. Note that this implies they are non-contractible, as the 0 element is never independent of anything, so in particular, they are not triangles.

⟨lem:hMV2⟩ **Lemma 7.7.** *Let A and B be disjoint isomorphic independent cycles in a graph G. Let E by a copy of $P_\ell \times A$ for $\ell \geq 4$, and let $G + E$ be the graph we get by identifying $\{0\} \times A$ in E with A and $\{\ell\} \times A$ in E with B. Let C be any cycle in $G + E$ whose intersection with E is a path with one endpoint in each of A and B. We have that*

$$H_1(\mathbf{G} + \mathbf{E}) = \langle (\mathcal{B} \setminus \{B\}) \cup \{C\} \rangle$$

for any basis of $H_1(\mathbf{G})$ containing $[B]$.

(Note that if $[B]$ is torsion-free, as it will be for us, then $H_1(\mathbf{G} + \mathbf{E}) \cong H_1(\mathbf{G})$.)

Proof. As the intersection of G and E is $A \cup B$, the sequence

$$\begin{aligned} \longrightarrow H_1(\mathbf{A} \cup \mathbf{B}) \quad &\xrightarrow{\Phi_1} \quad H_1(\mathbf{G}) \oplus H_1(\mathbf{E}) \xrightarrow{\Psi_1} H_1(\mathbf{G} + \mathbf{E}) \\ &\xrightarrow{\delta_1} \quad \tilde{H}_0(\mathbf{A} \cup \mathbf{B}) \xrightarrow{\Phi_0} \tilde{H}_0(\mathbf{G}) \oplus \tilde{H}_0(\mathbf{E}) \to \dots \end{aligned}$$

from (1), which quickly becomes

$$\to H_1(\mathbf{A} \cup \mathbf{B}) \xrightarrow{\Phi} H_1(\mathbf{G}) \oplus H_1(\mathbf{E}) \xrightarrow{\Psi} H_1(\mathbf{G} + \mathbf{E}) \xrightarrow{\delta} \mathbb{Z} \to 0,$$

is exact.

Here $\Phi : \alpha \mapsto (\alpha, -\alpha)$ and $\Psi : (\alpha, \beta) \mapsto \alpha + \beta$. So we see that, $\Psi(H_1(\mathbf{G}) \oplus H_1(E)) = \Psi(H_1(\mathbf{G}) \oplus \{0\})$. Indeed, $H_1(E)$ has only the one non-zero element $[A] = [B] =: 1$ and so for any $[C] \in H_1(\mathbf{G})$,

$$\Psi(([C], 1)) = \Psi([C], [A]) = [C] + [A] = \Psi([C] + [A], 0).$$

Thus the image of Ψ is the image of its restriction Ψ_r to $H_1(\mathbf{G}) \oplus \{0\}$. To determine this, we observe that the kernal of Ψ_r is, by exactness

$$\begin{aligned} \ker(\Psi) \cap (H_1(\mathbf{G}) \oplus \{0\}) &= \operatorname{Im}(\Phi) \cap (H_1(\mathbf{G}) \oplus \{0\}) \\ &= \{(\sigma, 0) \mid [\sigma] = 0 \text{ in } H_1(\mathbf{E})\} \\ &= \langle([A - B], 0)\rangle. \end{aligned}$$

So Ψ_r, and so Ψ injects $H_1(G)/\langle A - B\rangle$ into $H_1(G + E)$. On the other hand, by exactness, δ surjects onto $\mathbb{Z}$ so has co-rank of 1, so $H_1(\mathbf{G} + \mathbf{E})/\operatorname{Im}(\Psi_r) \cong \mathbb{Z}$. As Z is free, we get

$$H_1(\mathbf{G} + \mathbf{E}) \cong \operatorname{Im}(\Psi_r) \oplus \mathbb{Z} = \frac{H_1(\mathbf{G})}{\langle A - B\rangle} \oplus \mathbb{Z}.$$

The factor $H_1(\mathbf{G})/\langle A - B\rangle$ is generated by the basis $\mathcal{B} \setminus \{[B]\}$ and the factor $\mathbb{Z}$ by any cycle C of $G + E$ such that $\delta(C \cap E) = a + b$, independent of this basis, this is any C made from a walk in E concatenated with a walk in G having the same (but reversed) endpoints.

□

⟨lem:hjoin2⟩ **Lemma 7.8.** *Let G be a graph and A and B be independent, isomorphic cycles of G being distance at least 4 apart. Let G' be the graph we get from G by identifying A and B. Then $H_1(\mathbf{G}')$ has a basis consisting of any basis of G containing A, less the cycle A, and $[C]$ where P is a walk from a vertex a in A to a vertex b in B that are identified in G' and $C = P/\{a, b\}$ is we get by identifying a and b in P.*

Proof. Make G_1 from G by taking a new a copy E of $P_l \times A$ and identifying copies of A. The basis does not change. Now make G_2 by attaching the other end to B. We get the basis we want by Lemma 7.7. Now we contract E to a cycle, one edge at a time, using Lemma 7.3 to get G'.

□

This final lemma says that if we 'plug' a basis cycle C in a graph with a vertex v_C adjacent to all of its vertices, then C becomes contractible and falls out of the basis. (This is basically the proof of Lemma 7.6 so we should join these.)

⟨lem:plug⟩ **Lemma 7.9.** *Let C be a cycle in an* NT*-basis of G, and let G' be the graph we get from G by adding a new vertex v_C adjacent to all vertices in C. Then $H_1(\mathbf{G}') = H_1(\mathbf{G}) \setminus \{[C]\}$.*

Proof. Let W be the wheel in G' induced by $V(C) \cup \{v_C\}$. It is easy to see that $H_1(\mathbf{W}) = 0$, and so by (1), the sequence

$$\xrightarrow{\delta} H_1(\mathbf{C}) \xrightarrow{\Phi} H_1(G) \oplus 0 \xrightarrow{\Psi} H_1(\mathbf{G}') \to 0$$

is exact. So Ψ surjects $H_1(\mathbf{C})$ onto $H_1(\mathbf{G}')$. As $H_1(\mathbf{C})$ is clearly $\langle[C]\rangle$, and so the image of Φ is $\langle([C], 0)\rangle$, this is the kernel of Ψ. So

$$H_1(\mathbf{G}') \cong H_1(\mathbf{G})/\langle[C]\rangle.$$

□

7.3. **Computing NT-bases of S_s and $G^{(a)}$.** In this section we prove Theorem 7.1, which gives us an NT-basis of $G^{(a)}$. First we find an NT-basis of the sum gadget S_s constructed in Section 5.

⟨lem:SiBasis⟩ **Lemma 7.10.** *For any $s \geq 2$, S_s has an* NT*-basis consisting of any s of its $s+1$ end gadget-cycles: $A_0, A_1, \dots A_{s-1}$ and Z.*

Proof. Referring to Construction 5.2, in which we constructed S_s, we started with $\mathcal{P} = P_\ell \times B_*$ which by Lemma 7.2 has an NT-basis consisting of either of its end gadget-cycles.

In step (1) we take a quotient of one of these end gadget-cycles by subpaths. Applying Lemma 7.3 one edge at a time, we have that the resulting graph $S_s^{(1)}$ still has an NT-basis consisting of either of its end gadget-cycles.

In step (2) we construct $S_s^{(2)}$ from $G^{(a.1)}$ by pinching several vertices together in the other end gadget-cycle $0 \times B_*$. As $0 \times B_*$ can be assumed to be a NT-basis of $S_s^{(1)}$, we can apply Lemma 7.5 each time we do this, and conclude that the resulting cycles $A'_1, \dots, A'_s$ make up an NT-basis of $S_s^{(2)}$.

For each attachment, in step (3), of a path-of-cycles $F = P_1 \times C_*$ to A'_i, we apply Lemma 7.6, with H containing $C_H = A'_i$ as our gadget so far, to get that A_i can replace A'_i in an NT-basis. Thus the A_i make up an NT-basis of $S_s^{(3)}$.

Lemma 5.4 tells us that the end gadget cycle Z is independent of any $s-1$ of these A_i, so can replace any one of them in an NT-basis. □

We are now ready to prove the main theorem of the section, that $G^{(a)}$ has an NT-basis consisting of gadget-cycles the cycles from an NT-basis $\mathcal{B}_T$ of T_*.

Proof of Theorem 7.1. In Step (1) of Construction 6.1 we introduce a set of $3n+1$ disjoint copies of C_*. Indeed, letting $V(G) = [n]$ and, for $i \in [n]$, letting $\mathcal{B}^i = \{A_0^i, A_1^i, A_2^i\}$, we introduce the set

$$\mathcal{B} := \bigcup_{i \in [n]} \mathcal{B}^i \cup \{Z_*\}$$

of copies of C_*.

In Step (2) we construct a graph $G^{(a.2)}$, the end gadget-cycles of which are the cycles in the set $\mathcal{B}$, as follows. We start with a copy $G_1 := S_3^1$ of S_3 and rename its copy of Z as Z_*. For $i = 2, \dots, n$, we construct G_i from G_{i-1} by taking a new copy S_3^i of S_3 and identifying its copy of Z with Z_*. The resulting G_n is $G^{(a.2)}$.

Now, by Lemma 7.10, we can assume that the copy of Z is in an NT-basis of each S_3^i, so applying Lemma 7.6 when we construct G_{i+1} from G_i we get an NT-basis of G_{i+1} by adding any two of the cycles in $\mathcal{B}^{i+1}$ to our NT-basis of G_i. It follows that for any choice of two cycles in $\mathcal{B}^i$ for each i, there is an NT-basis of $G^{(a.2)}$ containing these cycles. At this point T_* is still a tree, so has an empty NT-basis $\mathcal{B}_T$. The following statement is therefore true:

For any two cycles C, C' in $\mathcal{B}$ there is an NT-basis $\mathcal{B}^*$ of $G^{(a.2)}$ with

$$\{C, C'\} \cup \mathcal{T} \subset \mathcal{B}^* \subset \mathcal{B} \cup \mathcal{B}_T. \tag{2}$$ eq:basis

In Steps (3) and (4) we construct $G^{(a)}$ from $G^{(a.2)}$ by taking several new copies of S_2 and attaching them to $G^{(a.2)}$ by identifying their copies of A_0 and A_1 with cycles of $G^{(a.2)}$ in $\mathcal{B}$. Applying the following claim each time we do this, we add a

new cycle of T_* to $\mathcal{B}_T$ and verify that (2) holds for the newly constructed graph in place of $G^{(a.2)}$. Thus the lemma is proved with the following claim.

Claim 7.11. *Let G be a graph and C and C' be disjoint copies of C_* in an* NT-*basis $\mathcal{B}$ of G. Let T be a connected subgraph of G with one vertex in each of C and C'. Let G' be the graph we get from G and a disjoint copy S of S_2 by identifying C with $A_0 \leq S$ and C' with $A_1 \leq S$ in such a way that the tree $T(S)$ of S has one leaf identified with the vertex of T in C and one other leaf the vertex of T in C'. Let T' be the union of T and $T(S)$.*

Where $\mathcal{B}_T$ is an NT-*basis of T, there is a cycle C_T in T' such that*

(1) $\mathcal{B}_T \cup \{C_T\}$ is an NT-*basis of T', and*
(2) $\mathcal{B} \cup \{C_T\}$ is an NT-*basis of G'.*

Proof. Starting with G and the copy S of S_2, first identify A_0 in S with the cycle C of G. Call this graph $G^{(1)}$. By Lemma 7.6, we get an NT-basis $\mathcal{B}^{(1)}$ of $G^{(1)}$ by removing C from $\mathcal{B}$ and adding the cycles A_1 and Z of S. The NT-basis of T has not changed, as we have just attached a tree $T(S)$ to some vertex.

Now construct G' by identifying the cycles A_1 and C' of $G^{(1)}$. This makes T' by identifying vertices of T that are distance at least 4 apart, so by Lemma 7.5 creates a new cycle C_T in T' such that $\mathscr{B}_T \cup \{C_T\}$ is an NT-basis of T'. By Lemma 7.8, we can get an NT-basis of G' from $\mathcal{B}^{(1)}$ by removing the cycle C' and adding the cycle C_T. This basis contains Z and A_1 of S. But in S, the cycles Z, A_1, and A_0 minimally dependent, so we can replace Z with $A_0 = C$ in this basis, getting that $\mathcal{B} \cup \{C_T\}$ is an NT-basis of G'. ◇

As mentioned before the claim, this completes the proof of the theorem. □

8. The full construction

⟨sect:finalproof⟩ Now we can finish our construction of the instance G^* of NonFlat(C_*) for an instance G of 3-colouring.

⟨const:G'⟩ **Construction 8.1.** *For a graph G, let $G^{(a)}$, with subgraph T_*, be given by Construction 6.1. By Theorem 7.1, $G^{(a)}$ has an* NT-*bases consisting of gadget cycles and cycles from a basis $\mathcal{B}_T$ of T_*. Construct the graph G^* from $G^{(a)}$ by adding, for each cycle C in $\mathcal{B}_T$, a new vertex v_C, and edges from v_C to each vertex of C.*

The following is an immediate corollary of Theorem 7.1 by Lemma 7.9.

⟨cor:topo⟩ **Corollary 8.2.** *For a graph G, the graph G^* constructed from $G^{(a)}$ in Construction 8.1 has an* NT-*basis consisting of end gadget-cycles of G^*.*

At last, we prove our main technical result, Lemma **??**, which says that for a reflexive cycle C_* of length at least 4, the problem NonFlat(C) is NP-complete.

Proof of Lemma **??**. For an irreflexive instance G of 3-colouring, let G^* be the graph from Construction 8.1. We show that G is 3-colourable if and only if G^* has a non-flat homomorphism to C_*.

Indeed, assume there is a non-flat C_*-colouring ϕ of G^*. By Fact **??**, some cycle in any NT-basis has non-zero wind, so by Corollary 8.2 some end gadget-cycle of $G^{(a)}$ has non-zero wind. This hold for the restriction of ϕ to $G^{(a)}$, and so by Lemma 6.6 Z_* does has non-zero wind. As is a copy of C_* the wind is ± 1 and we may assume that it is 1. By Lemma 6.4, we thus have that there is a 3-colouring of G.

On the other hand, assume that there is a 3-colouring of G. By Lemma 6.4 there is a non-flat C_*-colouring ϕ of $G^{(a)}$ which we may assume is 0 on T_*. Setting $\phi(v_C) = 0$ for every $C \in \mathcal{B}_T$ this extends to a non-flat C_*-colouring of G^*. □

By Lemma **??** and Corollary **??** we have thus proved the following main case of our main theorem.

?⟨thm:cycles⟩? **Theorem 8.3.** *For any reflexive cycle C of girth at least 4, $\mathrm{Mix}(C_*)$ is coNP-complete.*

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